



BRAC UNIVERSITY
Department of Computer Science & Engineering
FINAL EXAMINATION, Summer 2025
CSE 350: Digital Electronics and Pulse Techniques



Total Marks: 50

Time: 2 Hours

- There are **four(4)** questions. Answer them **all**. Question 4 is a **bonus** question.
- Figure in bracket [] next to each question indicates marks for that question.
- Write your set number on top of your answer sheet.
- $V_{BE}(\text{saturation}) = 0.8\text{ V}$, $V_{CE}(\text{saturation}) = 0.2\text{ V}$, $V_{BC}(\text{reverse active}) = 0.7\text{ V}$, $V_D(\text{forward bias}) = 0.7\text{ V}$ if not stated otherwise.

Name:	ID:	Section:
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Question 1:

[20]

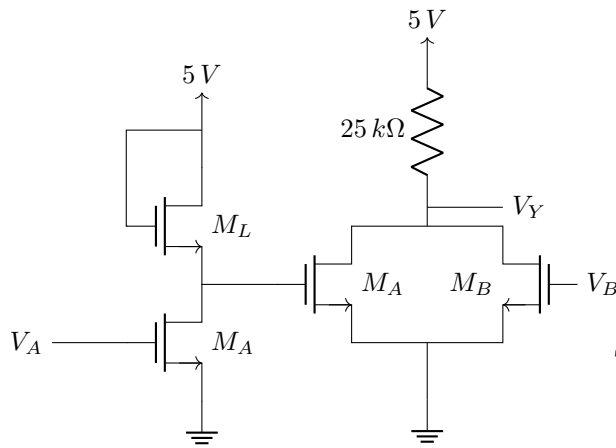


Figure 1

	$K_n \left(\frac{mA}{V^2} \right)$	$V_{TN} (V)$
M_L	0.3	1.4
M_A	0.6	0.6
M_B	0.7	0.4

Table 1: MOSFET parameters

Triode Mode: $I_{DS} = K_n[2(V_{GS} - V_{TN})V_{DS} - V_{DS}^2]$

Saturation Mode: $I_{DS} = K_n(V_{GS} - V_{TN})^2$

CO1	(a)	Write the Boolean function expression of the output (Y) in terms of the inputs (A & B).	[2]
CO1	(b)	Find the power dissipation of the full circuit when $V_A = V_B = 5\text{ V}$ (high).	[5]
CO1	(c)	If the $20\text{ k}\Omega$ was replaced with M_L , find the output voltage V_Y when, $V_A = 0\text{ V}$ (low) & $V_B = 5\text{ V}$ (high).	[5]

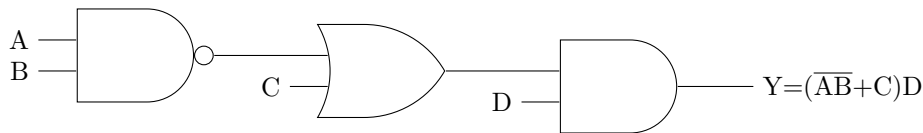


Figure 2

CO1	(d)	Implement the Boolean function shown in Figure 2 using CMOS logic .	[8]
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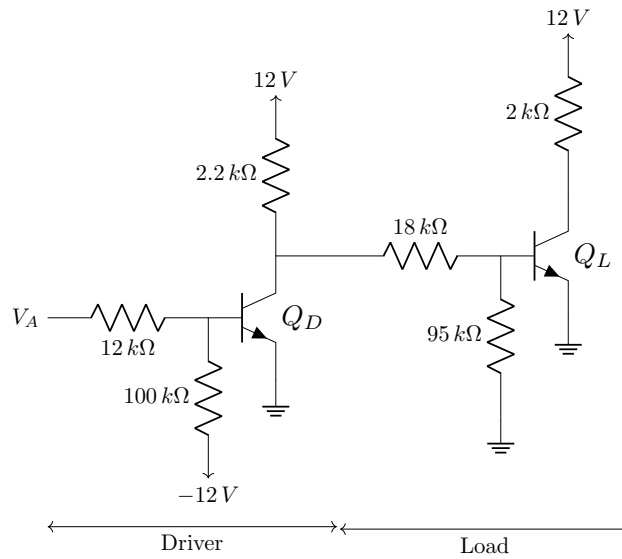


Figure 3

For the circuit shown in Figure 3, $\beta_F = 30$, $V_{OH} = 10\text{ V}$, $V_{OL} = 0.2\text{ V}$, high input = 12 V , low input = 0.2 V .

CO2	(a)	Find the noise margin of the circuit.	[5]
CO2	(b)	Find the maximum fanout of the driver.	[5]

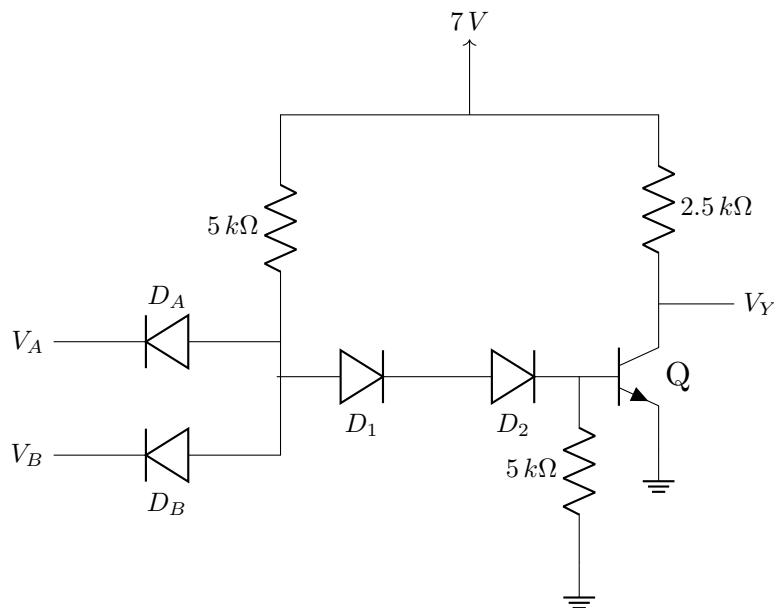


Figure 4

For the circuit shown in Figure 4, $\beta_F = 30$, $V_{OL} = 0.2\text{ V}$, high input = 7 V , low input = 0.2 V .

CO2	(c)	Find the average power dissipation of the circuit.	[5]
CO2	(d)	If the circuit in Figure 4 is used as both driver & load, find the maximum fanout .	[5]

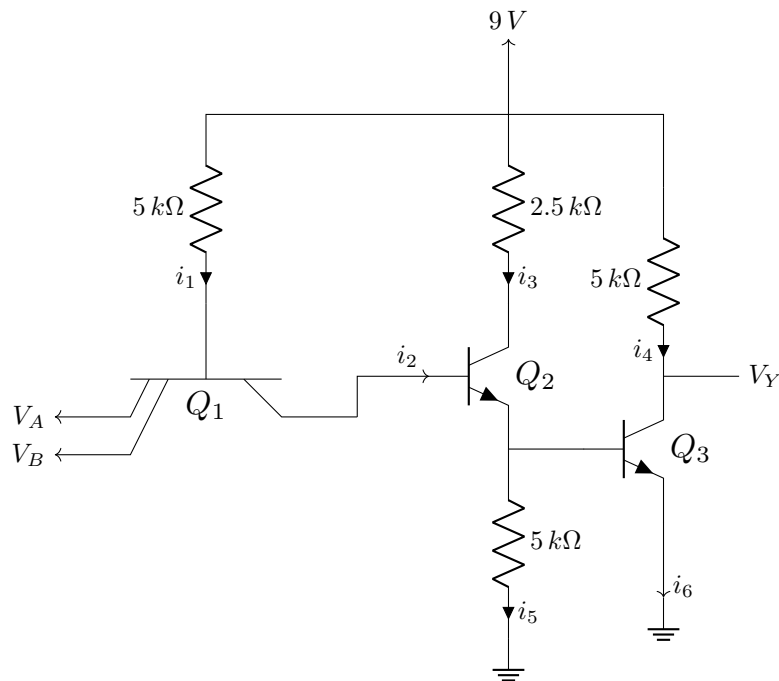


Figure 5

For the circuit shown in Figure 5, high input = 9 V, low input = 0.2 V.

CO1	(a)	If $i_6 = 6.81 \text{ mA}$, find the value of reverse beta , β_R of Q_1 .	[5]
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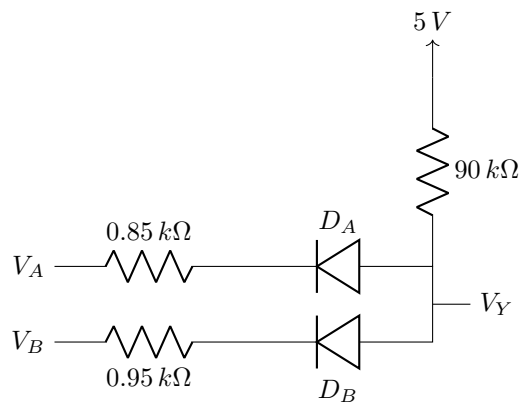


Figure 6

CO1	(b)	For the circuit in Figure 6, find the output voltage V_Y and the power dissipated when $V_A = \text{high}$ & $V_B = \text{low}$. [high input = 5 V, low input = 0 V]	[5]
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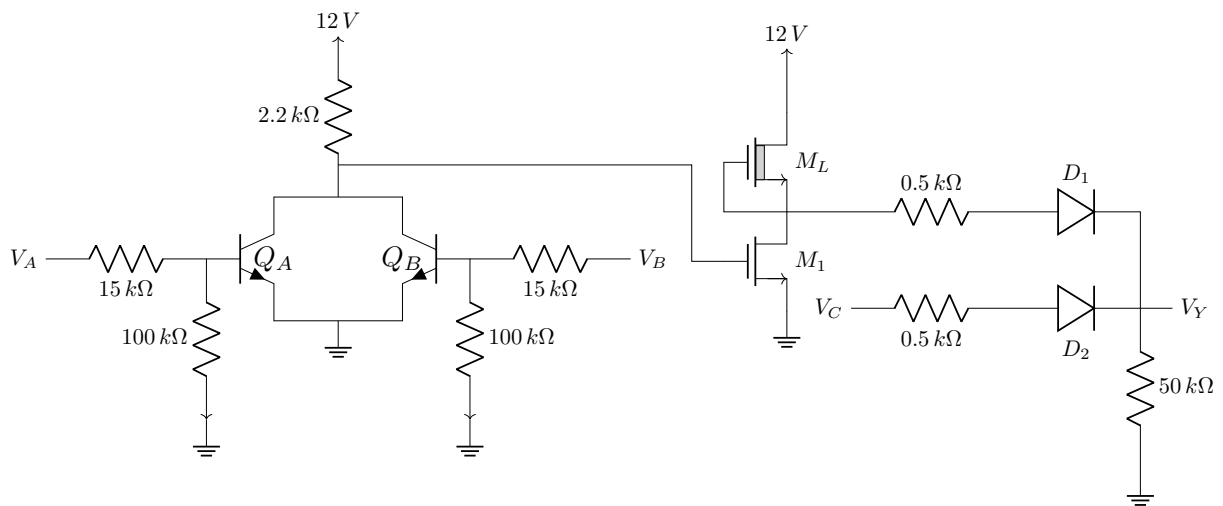


Figure 7

Fill in the following blanks (on the question paper) based on Figure 7—

CO1	(a)	Boolean function of the output: $Y = \text{-----}$	[1]
CO1	(b)	Minimum value of V_{TN} of M_1 for the inverter to work properly: $V_{TN_1} = \text{-----}$	[1]
CO1	(c)	If Q_B is in cut-off, D_2 is off & V_Y is low, operating mode of Q_A : -----	[1]
CO1	(d)	If V_Y is low, operating mode of M_1 : -----	[1]
CO1	(e)	If there were no restrictions on the number of inputs, which of the above gates alone can implement the same logic function? Ans: -----	[1]
CO1	(f)	If exactly two of the gates were to be used to implement 'Y', which would they be? Ans: ----- How?: -----	[2]
CO1	(g)	If $V_A = \text{high}$, operating mode of M_L : -----	[1]
CO1	(h)	Implement $F = \overline{Y}A + \overline{A}B$ using exactly one NMOS.	[2]